

TrendWatch AI — Social Media Intelligence Report

Use Case 2: Brand monitoring on Reddit · Generated 2026-06-09 03:38 UTC

Query: microsoft r/technology Limit: 48 Time: devvit_import

Data: Live Reddit (imported from Devvit app)

Report contents (assignment sections)

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1. Executive briefing

Source: rule_based_fallback · RAG + CoT LLM or rule-based fallback

Executive summary

TrendWatch AI analysed 48 Reddit posts. Negative sentiment is 35.4%. Crisis score is 50/100 (Medium).

Main concerns

- Risk keywords appeared 3 times across posts.
- 16 high-engagement posts carry negative sentiment.
- Negative sentiment ratio rose to 100% (average 37%). Monitor closely.

Crisis explanation

Risk keywords appeared 3 times across posts. 16 high-engagement posts carry negative sentiment. Negative sentiment ratio rose to 100% (average 37%). Monitor closely.

Recommended actions

- Monitor high-impact negative threads daily.
- Prepare a factual FAQ for top complaint topics.
- Escalate to customer support if refund/delay keywords spike.

Communication strategy

Address emerging topics: Microsoft / From / Openai, Microsoft / Windows / Chip, Microsoft / With / Windows. Use empathetic, transparent tone; avoid dismissive responses.

Full LLM report (markdown)

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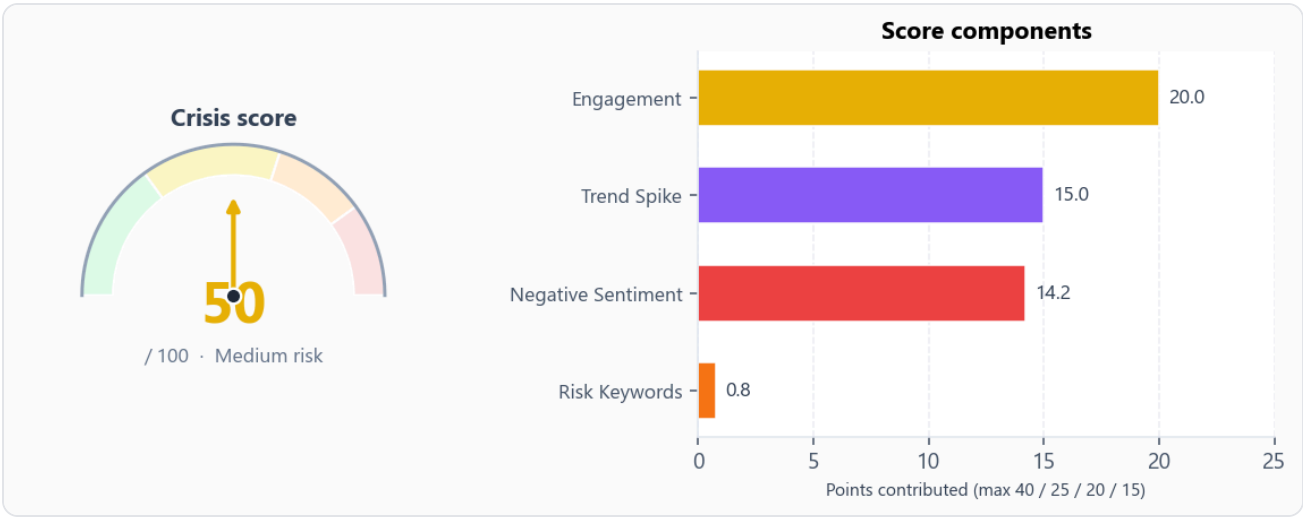
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2. Crisis intelligence layer



50/100 — Medium risk

Weighted fusion of negative sentiment, risk keywords, high-engagement negative posts, and trend spikes.

Score components

Component	Points
negative_sentiment	14.2
risk_keywords	0.8
engagement	20.0
trend_spike	15.0

Reasons

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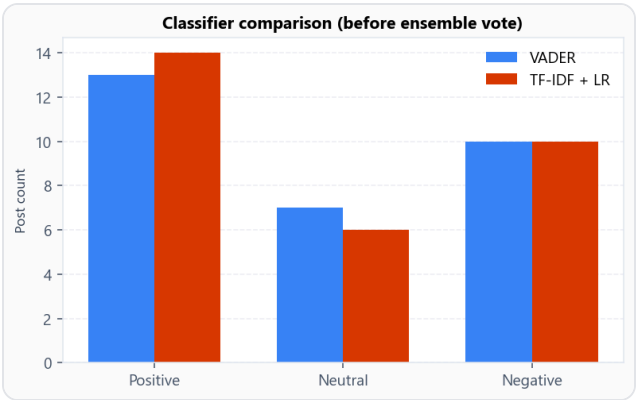
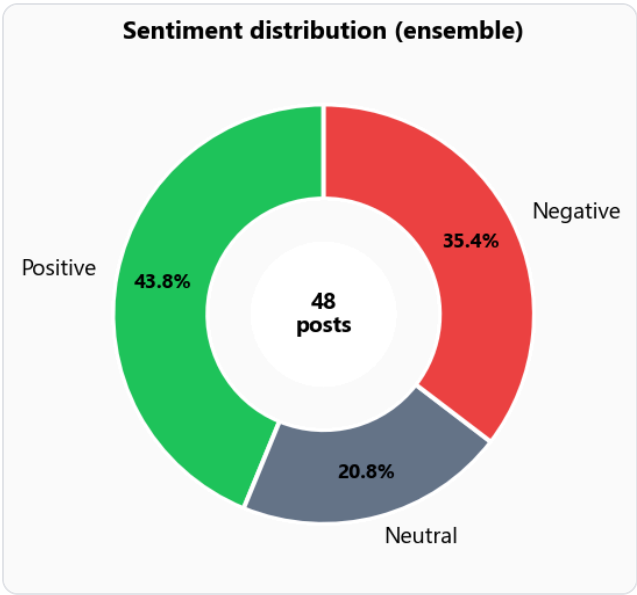
3. Overview & sentiment analysis

48
POSTS ANALYSED

43.8%
POSITIVE

20.8%
NEUTRAL

35.4%
NEGATIVE



Independent classifier outputs before ensemble fusion.

Final labels after VADER + LR ensemble majority vote.

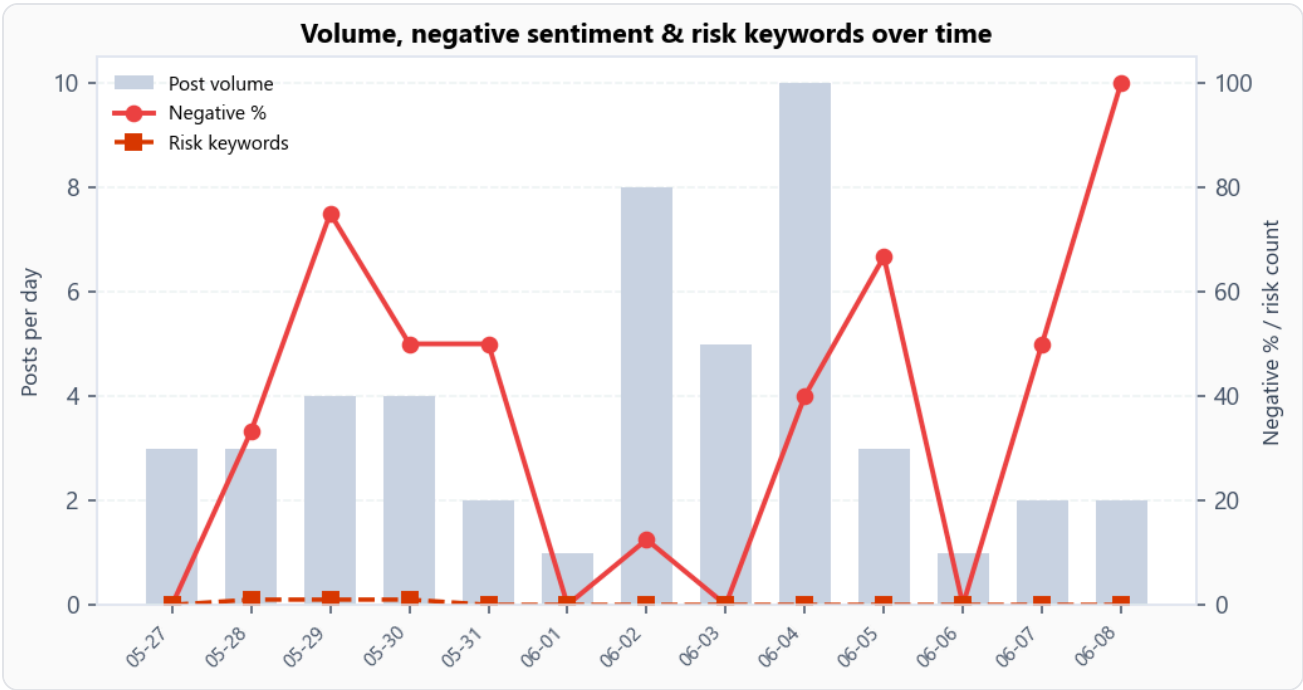
Sentiment distribution (VADER + LR ensemble)

Positive	43.8%
Neutral	20.8%
Negative	35.4%

Label	Count
positive	21
neutral	10
negative	17

Counts: positive=21, neutral=10, negative=17

4. Trend detection



Daily post volume (bars), negative sentiment % (line), and risk keyword counts (dashed).

Spike detected: Yes — Negative sentiment ratio rose to 100% (average 37%). Monitor closely.

Average negative ratio: 36.7%

Date	Posts	Negative	Neg %	Risk keywords
2026-05-27	3	0	0.0%	0
2026-05-28	3	1	33.3%	1
2026-05-29	4	3	75.0%	1
2026-05-30	4	2	50.0%	1
2026-05-31	2	1	50.0%	0
2026-06-01	1	0	0.0%	0
2026-06-02	8	1	12.5%	0
2026-06-03	5	0	0.0%	0
2026-06-04	10	4	40.0%	0

Date	Posts	Negative	Neg %	Risk keywords
2026-06-05	3	2	66.7%	0
2026-06-06	1	0	0.0%	0
2026-06-07	2	1	50.0%	0
2026-06-08	2	2	100.0%	0

5. Topic clusters (embeddings + KMeans)

Topic 1: Microsoft / From / Openai

Size: 18 · Avg sentiment: **positive**

Keywords: microsoft, from, openai, data, people, chief

Representative: Microsoft data suggests using AI is more expensive than hiring people

Topic 3: Microsoft / Windows / Chip

Size: 11 · Avg sentiment: **positive**

Keywords: microsoft, windows, chip, linux, build, killing

Representative: Euro-Office, Europe's open-source alternative to Microsoft Office and Google Docs, launches June 9

Topic 4: Microsoft / With / Windows

Size: 9 · Avg sentiment: **negative**

Keywords: microsoft, with, windows, after, browser, will

Representative: Microsoft Office 2019 for Mac will no longer edit documents after July 13

Topic 0: Microsoft / Surface / Laptop

Size: 5 · Avg sentiment: **positive**

Keywords: microsoft, surface, laptop, ultra, spark, nvidia

Representative: Microsoft builds its ultimate MacBook Pro rival with the NVIDIA powered Surface Laptop Ultra

Topic 2: Microsoft / Researcher / Zero

Size: 5 · Avg sentiment: **negative**

Keywords: microsoft, researcher, zero, exploits, action, company

Representative: Microsoft's GitHub bans security researcher who posted zero-day Windows exploits because company "ruined their life" — expert claims action is vindictive and promises further retaliation

6. Named entities (NER)

Entity	Mentions
microsoft (ORG)	27
11 (CARDINAL)	4
microsoft s (ORG)	4
congress (ORG)	2
devs (PERSON)	2
microsoft debuts (ORG)	2
nvidia (ORG)	2
microsoft data (ORG)	2
zero day (DATE)	2
pentagon (ORG)	2
9 7 billion (MONEY)	2
the 2nd (DATE)	1
weeks (DATE)	1
years (DATE)	1
satya nadella (PERSON)	1

7. Risk keywords & high-impact posts

Risk keyword frequency

Keyword	Count
exploit	3

Similar post pairs (cosine similarity)

Post A	Post B	Similarity
AI CEOs from OpenAI, Anthropic, and Microsoft set aside	AI CEOs from OpenAI, Anthropic, and Microsoft set aside	1.0
Microsoft data suggests using AI is more expensive than	Microsoft data suggests using AI is more expensive than	1.0
t3_1tpqikl	t3_1tpjbd9	1.0
The Nvidia RTX Spark Era Starts Here: Hands On With Mic	Microsoft debuts Surface RTX Spark Dev Box — Nvidia-pow	0.754
The Nvidia RTX Spark Era Starts Here: Hands On With Mic	t3_1ttlbc	0.687

High-impact posts

Title	Subreddit	Score	Comments	Sentiment	Risk terms	Impact
Microsoft data suggests using AI is more expensive than hiring people	technology	6326	326	neutral	—	14.54

Title	Subreddit	Score	Comments	Sentiment	Risk terms	Impact
Microsoft Wants to 'Make People Addicted' to its New AI Assistant, Internal Docu...	technology	4036	505	positive	—	14.53
Microsoft data suggests using AI is more expensive than hiring people	technology	5093	331	neutral	—	14.34
Microsoft Office 2019 for Mac will no longer edit documents after July 13	technology	1526	276	negative	—	12.96
Microsoft CEO says new AI data centers use as little water annually as a restaur...	technology	955	440	negative	—	12.95

8. Post-level analysis (sample)

Showing up to 30 posts with VADER label, LR label, and extracted risk terms.

Title	Author	Ensemble	VADER	LR	Risk terms
For the 2nd time in weeks, Microsoft packages laced with credential st...	lurker_bee	negative	negative	negative	—
Microsoft AI chief on why it's 'dangerous' to call AI 'alive'	serene_sketch	negative	negative	negative	—
Microsoft is letting you kill Bing in Windows 11 Search, after years o...	Quantum-Coconut	negative	negative	negative	—
AI CEOs from OpenAI, Anthropic, and Microsoft set aside their rivalry ...	Just-Grocery-2229	positive	positive	positive	—
Microsoft AI chief says company was "set free" from OpenAI to pursue s...	Logical_Welder3467	positive	positive	positive	—
Browser coalition accuses Microsoft of monopolistic abuse and demands ...	rkhunter_	negative	negative	negative	—
AI CEOs from OpenAI, Anthropic, and Microsoft set aside their rivalry ...	lKeepItLayingAround	positive	positive	positive	—
Satya Nadella 'not sure' who said Microsoft wanted to make addictive A...	marketrent	negative	negative	negative	—
Surface Laptop Ultra targets 110W TDP for RTX Spark Superchip — Micros...	Logical_Welder3467	positive	positive	positive	—
Microsoft blames unexpected Windows driver updates on caching issue	lurker_bee	negative	negative	negative	—

Title	Author	Ensemble	VADER	LR	Risk terms
Microsoft is ditching password-based authentication tomorrow – Edge br...	rkhunter_	neutral	neutral	neutral	—
Microsoft continues its big Linux push at Build 2026	CackleRooster	neutral	neutral	neutral	—
Researcher Drops a New VS Code Zero-Day After Losing Trust in Microsof...	CircumspectCapybara	positive	positive	positive	—
The Nvidia RTX Spark Era Starts Here: Hands On With Microsoft's Surfac...	Logical_Welder3467	positive	positive	positive	—
Microsoft Office 2019 for Mac will no longer edit documents after July...	venice_mcgangbang	negative	negative	negative	—
Microsoft CEO says new AI data centers use as little water annually as...	yourfavchoom	negative	negative	negative	—
Microsoft's Coreutils project brings Linux commands to Windows	rkhunter_	neutral	neutral	neutral	—
Microsoft is killing Windows 11's web app slop, encourages devs to bui...	Quantum-Coconut	negative	negative	negative	—
Microsoft launches Scout, an OpenClaw-inspired personal assistant	lurker_bee	positive	positive	positive	—
Microsoft claims new quantum chip 1,000 times better than before	Scary_Statement4612	positive	positive	positive	—
Microsoft's Big New Idea for AI Gadgets Is a Badge With a Camera	Logical_Welder3467	neutral	neutral	neutral	—

Title	Author	Ensemble	VADER	LR	Risk terms
'We need an AI that places humanity first': Microsoft AI CEO outlines ...	Logical_Welder3467	positive	positive	positive	—
Microsoft Targets Legal Fears to Sell Its Powerful New AI Model to Bus...	Logical_Welder3467	positive	positive	positive	—
Microsoft announces Majorana 2 quantum computing chip — claims a pract...	Logical_Welder3467	neutral	neutral	neutral	—
Microsoft debuts Surface RTX Spark Dev Box — Nvidia-powered mini-PC he...	Logical_Welder3467	positive	positive	positive	—
Microsoft teases new era of AI-driven devices at annual developer conf...	talkingatoms	negative	negative	negative	—
Microsoft debuts Scout agent, homegrown reasoning model	joe4942	neutral	neutral	neutral	—
Microsoft Wants to 'Make People Addicted' to its New AI Assistant, Int...	MarvelsGrantMan136	positive	neutral	positive	—
Microsoft insists Defender is enough for most PCs, but admits third-pa...	rkhunter_	positive	positive	positive	—
Nvidia and Microsoft Researchers Say AI Agents Don't Care About Safety...	MarvelsGrantMan136	positive	positive	positive	—

9. NLP techniques demonstrated (rubric)

Minimum **3 basic** + **3 advanced** techniques — implementation mapping for your written report.

Basic techniques

- **Preprocessing** (tokenize, stem, lemmatize, URL removal)
- **Named entity recognition** (spaCy / NLTK)
- **Part-of-speech tagging**
- **TF-IDF + Logistic Regression classification**
- **Sentiment ensemble** (VADER + LR majority vote)
- **Embedding-based topic clustering** (KMeans)
- **Regex information extraction** (hashtags, mentions, risk lexicon)

Advanced techniques

- **Foundation LLM** (OpenAI-compatible briefing)
- **Retrieval-augmented generation** (top-k posts)
- **Prompt engineering** (structured analyst template)
- **Chain-of-thought prompting**
- **LLM-as-judge / automated evaluation**
- **Ensemble methods** (sentiment + multi-signal crisis)
- **Agentic ReAct-style orchestration**
- **Ethics & fairness disclaimers**

10. Agent orchestration & evaluation

Agent status: completed

ReAct-style trace

Step	Thought	Action	Observation
1	Need Reddit posts about the brand/topic.	collect_posts	Starting data collection.
2	Collected 48 posts.	collect_posts	Received 48 posts.
3	Raw text must be normalized for NLP.	preprocess	Running tokenizer pipeline.
4	Preprocessing complete.	preprocess	48 posts prepared.
5	Extract hashtags, mentions, risk keywords.	extract	Regex IE.
6	Run NER and POS for entity-aware monitoring.	ner_pos	spaCy/NLTK.
7	Classify sentiment with VADER + LR ensemble.	sentiment	Ensemble vote.
8	Cluster topics with embeddings.	topics	KMeans clustering.
9	Compute temporal trends.	trends	Daily aggregation.
10	Score crisis risk from sentiment + engagement + trends.	crisis	Crisis formula.
11	Retrieve relevant posts for grounded LLM briefing (RAG).	rag	Query=microsoft
12	Generate executive briefing with CoT LLM.	briefing	LLM or fallback.

Automated evaluation

Sentiment agreement (VADER vs LR): 93.8% — VADER and LR agree on 45/48 posts (94%). Use manually labelled gold set for formal accuracy in your report.

Briefing judge score: 70/100 (error_fallback)

LLM judge unavailable; use manual rubric in report.

11. Ethics, limitations & references

--- *Ethics notice: Analysis uses public Reddit text. Sentiment and crisis scores are algorithmic estimates, not legal or PR advice. Models may reflect language bias; human review is required before brand actions.*

- Public Reddit text only; no private user data.
- Crisis scores are decision-support estimates, not legal or PR advice.
- Sentiment models can misclassify sarcasm and domain-specific language.
- For quantitative evaluation on labelled data, extend `backend/evaluation.py`.
- Full NLP technique reference: `docs/NLP_TECHNIQUES.md`

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